
A EVALUATION OF TREATMENTS FOR TEMPOROMANDIBULAR DISORDER, WITH AND WITHOUT COMORBID HEADACHE.

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ABSTRACT

This study analyzes a retrospective dataset of 70 patients with Temporomandibular Disorder (TMD), with and without comorbid headache, to evaluate treatment effectiveness and compare responses between groups. Pain intensity was recorded across up to four visits, and percent change from the first to the last visit was used as the primary outcome. Paired t -tests indicated significant pain reduction over time in both groups, with no significant difference in treatment effectiveness between TMD and TMD with headache (HATMD) patients. Linear regression models adjusting for age and sex were used to further examine associations, yielding consistent results with no significant effects of headache status or other covariates on percent change in pain intensity. The model showed limited explanatory power ($R^2 = 0.057$). Key limitations include the small sample size, limited clinical variables, and substantial missing data at later visits, particularly in the HATMD group. Future studies with larger samples, more comprehensive patient characteristics, and improved handling of missing longitudinal data are needed to better understand variability in treatment response.

1 Introduction

This study analyzes a retrospective dataset of patients diagnosed with Temporomandibular Disorder (TMD), with and without comorbid headache. The dataset includes 70 patients and captures patient-level demographics, pain intensity measurements across up to four clinical visits, with some missing observations for patient with headache at later visits. The primary objective is to compare treatment response between patients with TMD alone and those with TMD accompanied by headache (HATMD).

Pain intensity is recorded at each visit, and treatment effectiveness is summarized using the percent change in pain intensity from the first visit to the fourth visit. This measure provides a standard way to quantify improvement over time.

Descriptive analyses are performed using distribution-appropriate summaries in the next section. To evaluate pain trajectories over time, two complementary visualizations are constructed. A paired t -test comparing pain intensity at the first and last visit was used to evaluate treatment effectiveness within the TMD and HATMD groups. In addition, a paired t -test was conducted on the percent change in pain intensity to assess whether treatment effectiveness differed between the TMD and HATMD groups.

To further evaluate factors associated with treatment response, a linear regression model is used with percentage change in pain intensity as the outcome. The model includes headache status as the primary variable of interest and adjusts for potential confounders, including age and sex. Regression coefficients are used to quantify the direction and magnitude of association between these variables and treatment response.

2 Dataset Preprocessing Description and Descriptive Statistics

The original dataset consisted of separate records for TMD and HATMD across four visits. These records were consolidated into a unified dataset, and a binary indicator variable was constructed to categorize patients by headache status, where 0 denotes TMD only and 1 denotes HATMD.

To evaluate pain trajectories over time, visualizations are constructed. Figure 1 is a patient-level longitudinal plot displays individual pain intensity trajectories across four visits. To have better visualization, we applied slight jittering (1/200 for each patient) to reduce overlap, allowing for visualization of within-patient variability and overall trends.

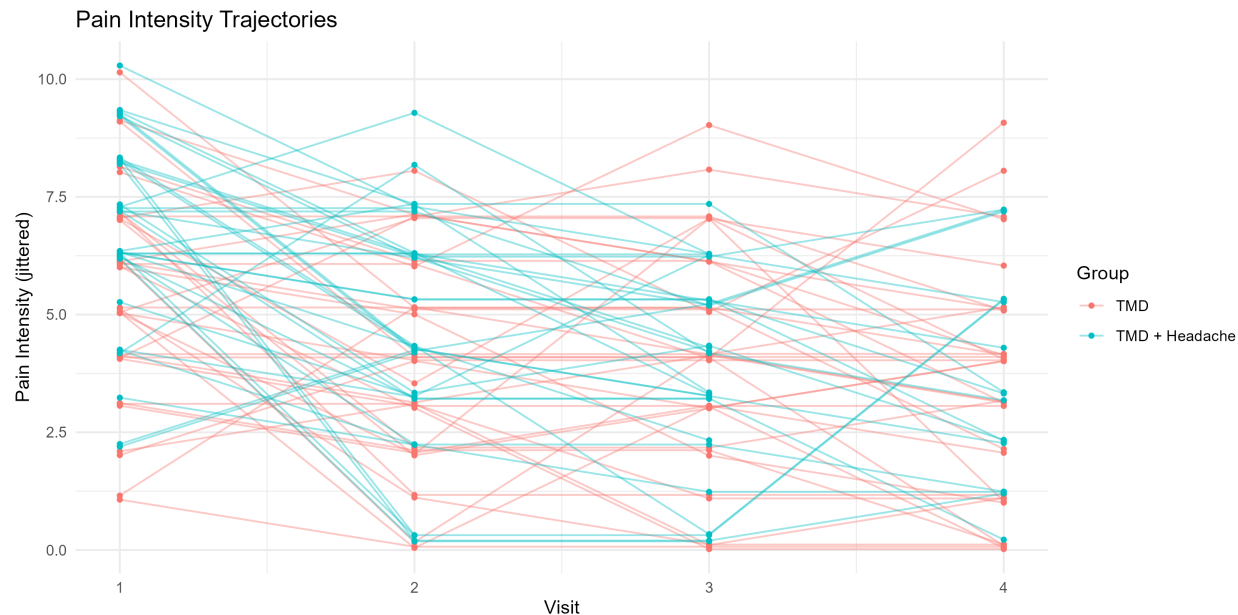


Figure 1: Pain intensity plot for both TMD and HATMD patients across four visits with 1/200 jittering

In Figure 1, the trajectories appear denser between the first two visits compared to later visits. This pattern likely reflects a higher proportion of missing observations at later visits, particularly among patients with TMD and headache, as shown in Table 1. Although pain intensity generally decreases over time, some patients exhibit increases at later visits. For instance, several TMD patients reported higher pain intensity at the third visit compared to the second. To better assess the overall trend, we can compare the second and third visits within the context of all four visits by fitting a line across the full set of observations for each patient.

Table 1: Missing Data Summary by Group (TMD vs. HATMD)

Variable	TMD (N = 35)	HATMD (N = 35)
Pain Intensity (Visit 3), no. missing	0	6
Pain Intensity (Visit 4), no. missing	0	18

To better visualize the pain intensity trend for TMD and HATMD group, we constructed a group-level summary plot in Figure 2. The figure shows the mean pain intensity at each visit for both groups, along with a 95% confidence interval error bars, providing a clear comparison of average pain progression over time. HATMD patients with missing visit was omitted in the mean and confidence intervals construction across visit in Figure 2.

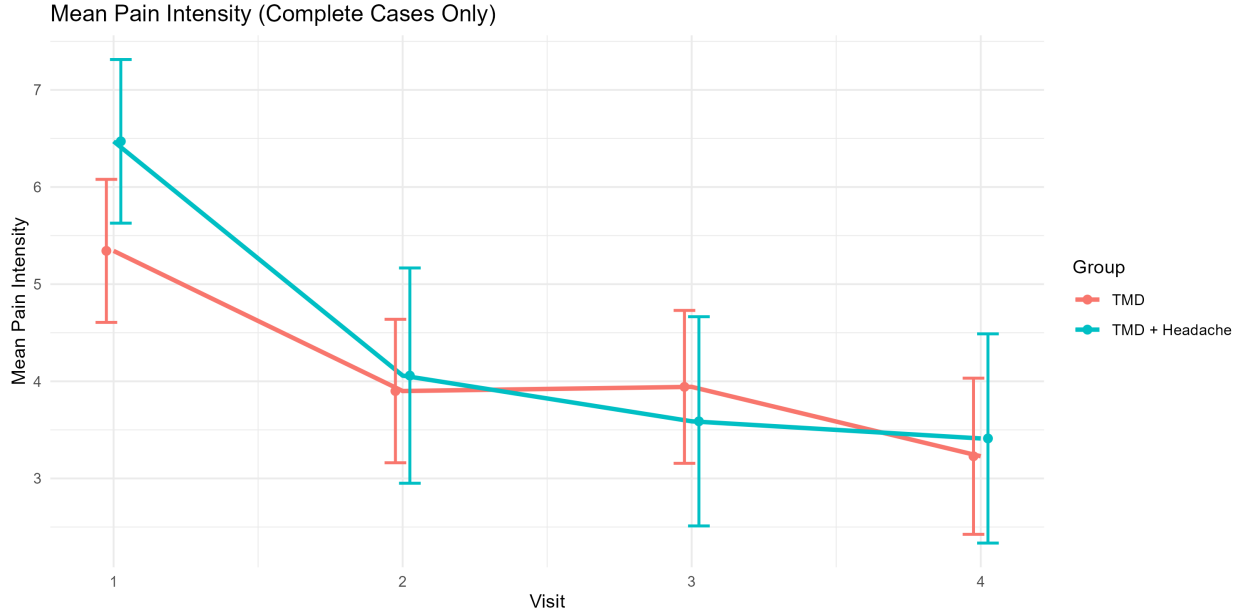


Figure 2: Mean pain intensity for both TMD and HATMD patients across four visits with 95% confidence interval error bars.

Both average curves in Figure 2 shows an overall decreasing trend which indicates the effectiveness of the treatments for both TMD and HATMD groups. The mean pain intensity for the TMD group increases in the third visit compared to the second, as seen the red line goes up in the third visit, which align with our initial observation from Figure 1. The wide error bars for the confidence interval is due to the small sample size. In particular, we observed a wider bar for the later visit of the HATMD group due to missing values.

To better quantify treatment effectiveness, we calculated the percentage change in pain intensity between the first (baseline) and fourth visit which is defined as:

$$\text{Percentage Change (\%)} = 100 \times \frac{\text{Pain Intensity (Visit 1)} - \text{Pain Intensity (Visit 4)}}{\text{Pain Intensity (Visit 1)}}.$$

Table 2 presents the descriptive characteristics of patients by headache status, with continuous variables summarized as mean, and categorical variables reported as counts and percentages. Differences between groups (TMD vs. HATMD) were assessed using Pearson's Chi-square test for categorical variables and independent two-sample *t*-tests for continuous variables, as appropriate. The reported *p*-values reflect distributional differences between the TMD and HATMD groups based on these tests.

A *p*-value represents the probability of observing a result at least as extreme as the one obtained, assuming that there is no true association between the variable and the outcome. Smaller *p*-values provide stronger evidence against the null hypothesis of no association. In practice, a *p*-value less than 0.05 is commonly considered statistically significant, suggesting that the observed difference is unlikely to be due to random variation alone, with smaller values indicating stronger evidence.

Table 2: Descriptive Characteristics by Group (TMD vs. HATMD)

Characteristic	TMD , N = 35	HATMD , N = 35	p-value	Test Label
Pain Intensity (Visit 1), mean	5	7	0.020*	1
Pain Intensity (Visit 2), mean	3.90	4.37	0.395	1
Pain Intensity (Visit 3), mean	4	4	0.606	1
Pain Intensity (Visit 4), mean	3	3	0.791	1
Percent Change (%), mean	32	48	0.238	1
Age, median (IQR)	43	37	0.214	1
Sex (Female), n (%)	25 (71%)	28 (80%)	0.577	2

TMD = Temporomandibular Disorder. Significance codes: '****' $p < 0.001$, '**' $p < 0.01$, '*' $p < 0.05$. Test Label: 1 = Independent Two-sample t; 2 = Pearson's Chi-Square.

Baseline (visit 1) pain intensity was significantly higher in the TMD with headache group compared to the TMD-only group ($p = 0.020$) which is considered to be strongly significant. If treatment effectiveness differed between the two groups, we would expect to observe a statistically significant difference in the percentage change in pain intensity. However, no significant differences were observed at later visits or in percentage change between groups. Additionally, age and sex distributions were similar between groups. Missing data, particularly at later visits, were more prevalent in the TMD with headache group.

3 Paired t-test

To test whether the differences of the pain intensity in Figure 1 and Figure 2 between the first and last visit are significant, we conducted the Paired t-test.

We first test the differences of the pain intensity in the TMD group under the following hypothesis:

$$H_0 : \text{TMD mean Pain in Visit 4} = \text{TMD mean Pain in Visit 1};$$

$$H_1 : \text{TMD mean Pain in Visit 4} < \text{TMD mean Pain in Visit 1}.$$

The very small p value ($p = 1.905 \times 10^{-5}$) provides strong evidence against the null (H_0) hypothesis, indicating that the mean pain intensity at the last visit is strongly significantly lower than at the first visit. This suggests a statistically significant reduction in pain intensity over the course of treatment for the TMD group with a mean difference of 2.

Similarly, we conducted the following to test the treatment for HATMD group:

$$H_0 : \text{HATMD mean Pain in Visit 4} = \text{HATMD mean Pain in Visit 1};$$

$$H_1 : \text{HATMD mean Pain in Visit 4} < \text{HATMD mean Pain in Visit 1}.$$

The very small p value ($p = 1.286 \times 10^{-5}$) also provides strong evidence that the mean pain intensity at the last visit is significantly lower than at the first visit. This suggests a statistically significant reduction in pain intensity over the course of treatment for the HATMD group with a mean difference of 4.

We can also use Paired t test to test whether the effectiveness of the treatment in the TMD and HATMD group differs by testing:

$$H_0 : \text{HATMD mean percent change} = \text{TMD mean percent change};$$

$$H_1 : \text{HATMD mean percent change} \neq \text{TMD mean percent change}.$$

The resulting p value ($p = 0.3928$), which is greater than the conventional significance threshold of 0.05, provides no evidence to reject the null hypothesis. This suggests that there is no statistically significant difference in mean percent change in pain intensity between the TMD and HATMD groups, indicating similar treatment effectiveness across the two groups.

4 Linear Regression

To examine the relationship between pain intensity percent change and pain group (TMD vs. HATMD), while accounting for patient demographic characteristics, we employed regression models. Specifically, we used linear regression, as the percent change in pain intensity is a continuous numeric outcome.

The goal is to isolate the independent relationship between each variable and the outcome, while accounting for the presence of other variables. This is particularly important in clinical data, where many factors are interrelated. By using these methods, we can better identify which variables are most strongly associated with the pain intensity percent change.

We also report p-values under the null hypothesis of 0 coefficient for the variable, which help assess whether an observed relationship is likely to be real or could have occurred by chance. A p-value in regression model represents the probability of observing a result at least as strong as the one found in the data, assuming that there is actually no true relationship between the variable and the outcome.

Smaller p-values indicate stronger evidence against the idea that there is no relationship. In practice, a p-value less than 0.05 is commonly considered statistically significant, meaning the observed association is unlikely to be due to random variation alone.

The result for fitting the linear model:

$$\text{Percent Change} \sim \text{Age} + \text{Headache} + \text{Sex}$$

is reported in Table 3.

Table 3: Linear regression results for percent change in pain intensity: no variables are significant.

	Estimate	Std. Error	t value	p-value
(Intercept)	1.51	23.78	0.06	0.950
Age	0.44	0.41	1.07	0.292
Headache	14.68	16.46	0.89	0.377
Sex	16.93	17.81	0.95	0.346

Multiple $R^2 = 0.057$, Adjusted $R^2 = -0.002$. 18 observations were removed due to missing data.

The regression model from Table 3 shows limited explanatory power and does not identify any statistically significant predictors of percent change in pain intensity.

Overall goodness of fit of our linear models is assessed by R^2 , ranges between 0 and 1. It is the coefficient of determination, defined as the proportion of variance in the outcome explained by the model. A value close to zero indicates that the model explains very little of the observed variability. The model has a very low coefficient of determination, with $R^2 = 0.057$ (adjusted $R^2 \approx 0$). This indicates that the included covariates (Age, Headache status, and Sex) explain only about 5.7% of the variability in the outcome. In practical terms, most of the variation in pain improvement remains unexplained by these baseline characteristics, suggesting poor model fit and weak predictive performance. This suggests that the relationship between the predictors and the outcome may be weak, nonlinear, or influenced by unmeasured factors.

None of the predictors are statistically significant at the 0.05 level (Age: $p = 0.292$, Headache: $p = 0.377$, Sex: $p = 0.346$). These large p-values indicate insufficient evidence to reject the null hypothesis that the corresponding regression coefficients are equal to zero. In other words, there is no strong statistical evidence that Age, Headache status, or Sex are associated with percent change in pain intensity in this dataset.

Since the factor Headache is also not statistically significant, this further supports our initial findings from the Paired t -test, which suggested that treatment effectiveness—measured by the percent change in pain intensity between the first and last visit—is similar between the TMD and HATMD groups. In particular, the lack of a significant group effect indicates that headache status does not appear to be associated with differential treatment response in this cohort.

5 Conclusion

This study evaluated treatment effectiveness in patients with TMD and HATMD, using longitudinal pain intensity data. Across both groups, Paired t -tests demonstrated a statistically significant reduction in pain intensity from the first to the last visit, indicating that treatment was effective in reducing pain over time for both TMD and HATMD patients.

When comparing treatment response between groups, no statistically significant difference was observed in percent change in pain intensity. This finding was consistent across both the Paired t -test and the linear regression analysis, where headache status was not a significant predictor of treatment response. Additionally, none of the demographic variables (Age or Sex) showed a significant association with percent change in pain intensity. The regression model also

demonstrated limited explanatory power ($R^2 = 0.057$), suggesting that the included covariates explain only a small proportion of variability in treatment response.

Overall, the results suggest that while patients in both groups experience meaningful improvement over time, there is no evidence of differential treatment effectiveness between TMD and HATMD patients based on the available data.

6 Limitations

This study has several important limitations. First, the sample size is relatively small ($n = 70$), which limits statistical power and reduces the ability to detect potentially meaningful differences between groups. A larger sample size would improve the stability of estimates and increase the likelihood of identifying significant predictors of treatment response.

Second, the dataset includes limited patient-level characteristics. Only basic demographic variables such as age and sex were available, which restricts the ability to adjust for important confounders (e.g., medical history, demographic). The absence of richer information likely contributes to the weak explanatory performance of the regression model.

Third, there is substantial missing data at later visits, particularly in the HATMD group. This introduces potential bias in longitudinal comparisons and reduces the reliability of estimated trajectories over time. The missingness is not explicitly modeled and may not be random.

Finally, the analysis assumes linear relationships and does not explore potential nonlinear effects or interactions between variables, which may limit the ability to capture more complex patterns in treatment response.

Future studies with larger sample sizes, more comprehensive variables, and improved handling of missing longitudinal data would likely yield more robust and informative conclusions regarding treatment effectiveness in TMD and HATMD populations.

AI Use

AI tools (Chatgpt) were used to assist in refining the clarity, structure, and readability of this report. AI was used to assist in Latex to create Tables. I also consult AI for unfamiliar packages in R.